

RET fusion-hotspot benchmark report

Study: thca_tcga_pan_can_atlas_2018

Abstract

This report analyzes RET gene fusions from the thca_tcga_pan_can_atlas_2018 study, covering 36 fusion events. 35/36 fusions (97.2%) were in-frame. 33/36 fusions (91.7%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0833333$, not statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 97.2% in-frame and 91.7% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 36 analyzed fusion events, identifying 10 distinct fusion partner genes. The most frequent partner was CCDC6, observed in 22 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 33/35 (94.3%) of in-frame fusions retained the domain, $p=0.0833333$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Disruption of the Cadherin domain was tested with Fisher's exact test comparing in-frame fusions against all others; 33/35 (94.3%) of in-frame fusions disrupted the domain, $p=0.0833333$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.436356.

Cutpoint detection

Cutpoint detection scanned 36 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 712 aa (exon 11; permutation-corrected $p=0.000999001$, statistically significant at $\alpha=0.05$). This is -13 aa from the nearest configured domain boundary at 725 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=33$) and "not_retained" ($n=3$). For Frame_status, retained: 33/33 (100.0%, 95% CI 89.6%-100.0%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, $p=unavailable$.

Composite evidence score

Composite evidence score produced results for this run (10 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association produced results for this run; see the accompanying data for details.

Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

Window detection

Window detection produced results for this run (6 row(s) in window_scan, 2 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CCDC6	22	0.611111111111112	0.30004340774793187	0.036015868536055845	0.9631016042780749	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.47191811111744875	1
NCOA4	5	0.138888888888889	0.30004340774793187	0.036015868536055845	0.9905882352941177	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3343744391031885	2
ERC1	2	0.0555555555555555	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3090214979267179	3
AKAP13	1	0.0277777777777777	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30068816459338454	4
DLG5	1	0.0277777777777777	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30068816459338454	5
FKBP15	1	0.0277777777777777	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30068816459338454	6
SPECC1L	1	0.0277777777777777	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30068816459338454	7
TBL1XR1	1	0.0277777777777777	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30068816459338454	8
TRIM27	1	0.0277777777777777	0.30004340774793187	0.036015868536055845	0.9882352941176471	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.30068816459338454	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
MRLN	1	0.02777777777777776	0.30004340774793187	0.036015868536055845	0.0	0.6190551149821004	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.15245287047573752	10

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
627	0	33	1	2	0.0	0.08333333333333333	1.0791812460476249
663	1	32	1	2	0.0625	0.1619047619047619	0.7907403776916642
712	1	32	3	0	0.0	0.0005602240896358541	3.2516382204482124

domain_disruption tables

frame_domain_contingency_table

33	0
2	1

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
cadherin	Cadherin domain	36	3	0	33	0.0

domain_retention tables

frame_domain_contingency_table

33	0
2	1

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	36	33	0	3	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
12	33	36	0.9166666666666666

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AKAP13	1
CCDC6	22
DLG5	1
ERC1	2
FKBP15	1
MRLN	1
NCOA4	5
SPECC1L	1
TBL1XR1	1
TRIM27	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
512	712	200	4	32	1	32	0.0	0.0005602240896358541	3.2516382204482124	EVT-TCGA-CE-A13K-01-32, EVT-TCGA-CE-A482-01-33, EVT-TCGA-DJ-A4V5-01-31, EVT-TCGA-EL-A3ZP-01-21
612	712	100	4	32	1	32	0.0	0.0005602240896358541	3.2516382204482124	EVT-TCGA-CE-A13K-01-32, EVT-TCGA-CE-A482-01-33, EVT-TCGA-DJ-A4V5-01-31, EVT-TCGA-EL-A3ZP-01-21
713	738	25	32	4	32	1	None	0.0005602240896358542	3.251638220448212	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-5, EVT-TCGA-CE-A13K-01-10, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-29, ... (32 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
713	763	50	32	4	32	1	None	0.0005602240896358542	3.251638220448212	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-5, EVT-TCGA-CE-A13K-01-10, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-29, ... (32 total)
713	813	100	32	4	32	1	None	0.0005602240896358542	3.251638220448212	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-5, EVT-TCGA-CE-A13K-01-10, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-29, ... (32 total)
713	913	200	32	4	32	1	None	0.0005602240896358542	3.251638220448212	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-5, EVT-TCGA-CE-A13K-01-10, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-29, ... (32 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
612	712	100	4	32	1	32	0.0	0.0005602240896358541	3.2516382204482124	EVT-TCGA-CE-A13K-01-32, EVT-TCGA-CE-A482-01-33, EVT-TCGA-DJ-A4V5-01-31, EVT-TCGA-EL-A3ZP-01-21
713	738	25	32	4	32	1	None	0.0005602240896358542	3.251638220448212	EVT-TCGA-BJ-A0ZJ-01-2, EVT-TCGA-BJ-A28Z-01-5, EVT-TCGA-CE-A13K-01-10, EVT-TCGA-CE-A481-01-36, EVT-TCGA-CE-A482-01-29, ... (32 total)

Per-event results (results.tsv)

event_id	sample_id	position_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_signal_effect_on_frame
EVT-TCGA-FK-A3SG-01-1	TCGA-FK-A3SG-01		AKAP13-RET	AKAP13	in-frame	threonine_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA AKAP13-RET Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-BJ-A0ZJ-01-2	TCGA-BJ-A0ZJ-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-EM-A3AN-01-3	TCGA-EM-A3AN-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-FK-A4UB-01-4	TCGA-FK-A4UB-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-BJ-A28Z-01-5	TCGA-BJ-A28Z-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-FK-A3SE-01-6	TCGA-FK-A3SE-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-ET-A40T-01-7	TCGA-ET-A40T-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-ET-A3DR-01-8	TCGA-ET-A3DR-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-J8-A4HW-01-9	TCGA-J8-A4HW-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-CE-A13K-01-10	TCGA-CE-A13K-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A3V3-01-11	TCGA-DJ-A3V3-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A4UQ-01-12	TCGA-DJ-A4UQ-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A4V5-01-13	TCGA-DJ-A4V5-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-ET-A39R-01-14	TCGA-ET-A39R-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-DO-A1JZ-01-15	TCGA-DO-A1JZ-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-E3-A3E0-01-16	TCGA-E3-A3E0-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-E8-A44M-01-17	TCGA-E8-A44M-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3CY-01-18	TCGA-EL-A3CY-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-DJ-A2Q1-01-19	TCGA-DJ-A2Q1-01		CCDC6-RET	CCDC6	in-frame	there_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame

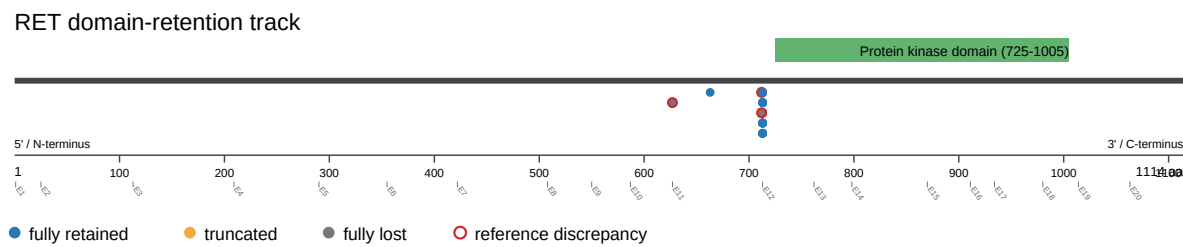
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-EL-A3TB-01-20	TCG A-EL-A3TB-01		CCDC6-R ET	CCDC6	in-frame	there _prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3ZP-01-21	TCG A-EL-A3ZP-01		CCDC6-R ET	CCDC6	in-frame	there _prime	43114587	11	663	False	663	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-EL-A3ZS-01-22	TCG A-EL-A3ZS-01		CCDC6-R ET	CCDC6	in-frame	there _prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA CDC6-RET Fusion	in-frame
EVT-TCGA-DE-A69K-01-23	TCG A-DE-A69K-01		DLG5-R ET	DLG5	in-frame	there _prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA DLG5-RET Fusion	in-frame
EVT-TCGA-EL-A3T9-01-24	TCG A-EL-A3T9-01		ERC1-RET	ERC1	in-frame	there _prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA ERC1-RET Fusion	in-frame
EVT-TCGA-EM-A3FQ-01-25	TCG A-E M-A3FQ-01		ERC1-RET	ERC1	in-frame	there _prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA ERC1-RET Fusion	in-frame

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-ET-A3DQ-01-26	TCGA-ET-A3DQ-01		FKBP15-RET	FKBP15	in-frame	threonine_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA FKBP15-RET Fusion	in-frame
EVT-TCGA-EM-A2CU-01-27	TCGA-EM-A2CU-01		NCOA4-RET	NCOA4	in-frame	threonine_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA NCOA4-RET Fusion	in-frame
EVT-TCGA-DE-A3KN-01-28	TCGA-DE-A3KN-01		NCOA4-RET	NCOA4	in-frame	threonine_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA NCOA4-RET Fusion	in-frame
EVT-TCGA-CE-A482-01-29	TCGA-CE-A482-01		NCOA4-RET	NCOA4	in-frame	threonine_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA NCOA4-RET Fusion	in-frame
EVT-TCGA-EL-A3H3-01-30	TCGA-EL-A3H3-01		NCOA4-RET	NCOA4	in-frame	threonine_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA NCOA4-RET Fusion	in-frame
EVT-TCGA-DJ-A4V5-01-31	TCGA-DJ-A4V5-01		RET-CCDC6	CCDC6	in-frame	five_prime	43114736	11	712	False	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		NA RET-CCDC6 Fusion	in-frame

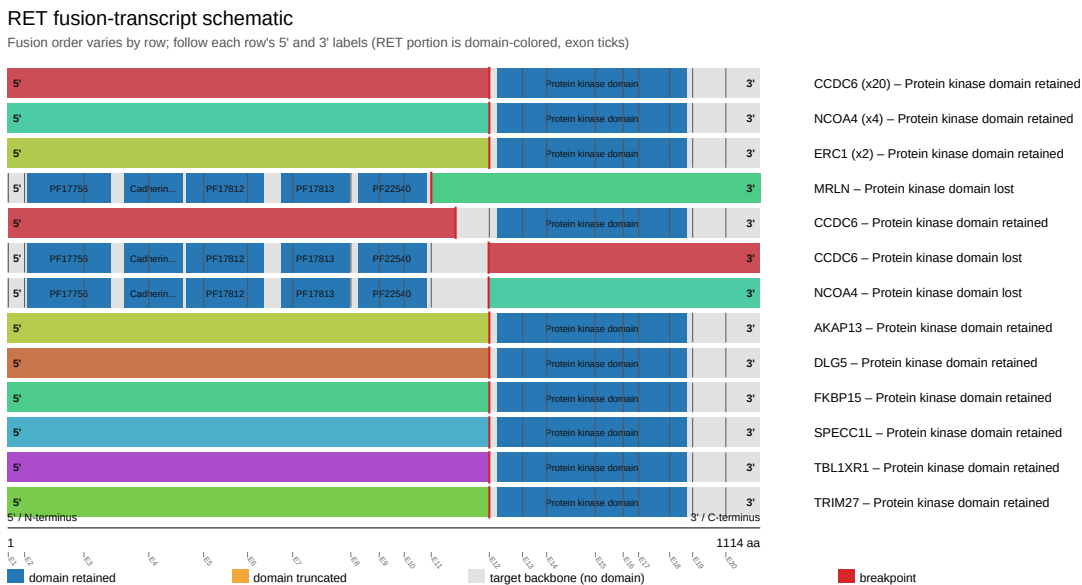
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-CE-A13K-01-32	TCGA-CE-A13K-01		RET-MRLN	MRLN	unknown	five_prime	43113675	10	627	False	626	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Cadherin domain	Protein kinase domain		NA RET-MRLN Fusion	NA
EVT-TCGA-CE-A482-01-33	TCGA-CE-A482-01		RET-NCOA4	NCOA4	in-frame	five_prime	43114736	11	712	False	712	True	exact match	lost	0.0	False	Cadherin domain	Protein kinase domain		NA RET-NCOA4 Fusion	in-frame
EVT-TCGA-CE-A485-01-34	TCGA-CE-A485-01		SPREC1L-RET	SPREC1L	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA SPECC1L-RET Fusion	in-frame
EVT-TCGA-ET-A40R-01-35	TCGA-ET-A40R-01		TBLL1XR1-RET	TBLL1XR1	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA TBL1XR1-RET Fusion	in-frame
EVT-TCGA-CE-A481-01-36	TCGA-CE-A481-01		TRIM27-RET	TRIM27	in-frame	three_prime	43116584	12	713	False	713	True	exact match	retained	1.0	False	Protein kinase domain	Cadherin domain		NA TRIM27-RET Fusion	in-frame

Figures

domain retention outliers



fusion schematic



reference comparison

Reference vs thca_tcga_pan_can_atlas_2018



